

Section : Common

Q.1

Match List-I with List-II

List-I	List-II
Function $f(x)$	Interval
(A) $f(x) = x x $	(I) Decreases on $(0, \infty)$
(B) $f(x) = x^2 + 2x - 5$	(II) Increases on $(3, \infty)$
(C) $f(x) = x^2 - 6x + 9$	(III) Decreases on $(-\infty, -1)$
(D) $f(x) = -x^2$	(IV) Increases on $(-\infty, \infty)$

Choose the correct answer from the options given below:

- (A) - (II), (B) - (III), (C) - (IV), (D) - (I)
- (A) - (IV), (B) - (III), (C) - (II), (D) - (I)
- (A) - (II), (B) - (IV), (C) - (III), (D) - (I)
- (A) - (II), (B) - (III), (C) - (IV), (D) - (IV)

Options 1. 1

2. 2

3. 3

4. 4

Q.2 Match List-I with List-II

List-I	List-II
Differential Equations	Degree/Order
(A) Degree of the differential equation $\frac{d^3y}{dx^3} + 2 \log x \cdot y = 0$	(I) 3
(B) Order of the differential equation $\frac{d^4y}{dx^4} + \left(\frac{dy}{dx}\right)^4 + xy = 0$	(II) 2
(C) Degree of the differential equation $\left(\frac{d^4y}{dx^4}\right)^2 + \left(\frac{dy}{dx}\right)^3 + x^2y = 0$	(III) 1
(D) Order of the differential equation $\frac{d^3y}{dx^3} + y\left(\frac{dy}{dx}\right)^3 = 0$	(IV) 4

Choose the correct answer from the options given below:

- (A) - (III), (B) - (II), (C) - (IV), (D) - (I)
- (A) - (II), (B) - (III), (C) - (IV), (D) - (I)
- (A) - (III), (B) - (IV), (C) - (II), (D) - (I)
- (A) - (II), (B) - (I), (C) - (IV), (D) - (III)

Options 1. 1

2. 2

3. 3

4. 4

Q.3 The maximum value of a LPP $z = 3x + 4y$ subject to the constraints: $x + y \leq 6$, $x \geq 0$, $y \geq 0$ is:

- 24
- 30
- 36
- 28

Options 1. 1

2. 2

3. 3

4. 4

Q.4 If $f(x) = 2x^3 - 15x^2 + 36x + 1$, $x \in [1, 5]$, then the absolute minimum value of $f(x)$ is:

1. 22

2. 28

3. 24

4. 29

Options 1. 1

2. 2

3. 3

4. 4

Q.5 Let x denote the number of heads in a simultaneous toss of three coins, then $P(0 < x \leq 3)$

1. $\frac{1}{2}$

2. $\frac{3}{4}$

3. $\frac{7}{8}$

4. 1

Options 1. 1

2. 2

3. 3

4. 4

Q.6 The particular solution of the differential equation $\frac{dy}{dx} = 8yx$ when $y = 1$ at $x = 0$ is:

1. $y = e^{x^2}$

2. $y = e^{4x^2}$

3. $y = e^{8x^2}$

4. $y = e^x$

Options 1. 1

2. 2

3. 3

4. 4

Q.7 The system of equations $2x + \lambda y = 8$, $\lambda x + 8y = 3$ has a unique solution if the value of λ is (are):

1. $\lambda \neq \pm 4$

2. $\lambda = 4$

3. $\lambda = -4$

4. λ is any real number

Options 1. 1

2. 2

3. 3

4. 4

Q.8 The area (in sq. units) of the region bounded by $y = -2$, $y = 2$, $x = y^3$ and $x = 0$ is equal to

1. 4

2. 6

3. 8

4. 16

Options 1. 1

2. 2

3. 3

4. 4

Q.9

The value of $\int_{-1}^1 |x| dx$ is

1. 0

2. 1

3. $\frac{1}{2}$

4. 2

Options 1. 1

2. 2

3. 3

4. 4

Q.10

If $z = 5x + 8y$ is the objective function of a LPP and $(0, 0), (3, 1), (2, 4), (0, 3), (5, 0)$ are corner points of the bounded feasible region, then the maximum value of the objective function is

1. 24

2. 42

3. 100

4. 23

Options 1. 1

2. 2

3. 3

4. 4

Q.11

$\int (e^{x \log a} + e^{a \log x}) dx$ is equal to (where $a > 1$)

1. $\frac{a^x}{\log a} + \frac{x^{a+1}}{a+1} + C$: C is an arbitrary constant

2. $(\log a) a^x + \frac{x^{a+1}}{a+1} + C$: C is an arbitrary constant

3. $\frac{a^x}{a+1} + \frac{x^{a+1}}{\log a} + C$: C is an arbitrary constant

4. $(a+1)a^x + (\log a)x^{a+1} + C$: C is an arbitrary constant

Options 1. 1

2. 2

3. 3

4. 4

Q.12

If $A = \begin{bmatrix} -1 & 2 & 3x \\ 2y & 4 & -1 \\ 6 & -1 & 0 \end{bmatrix}$ is a symmetric matrix, then the value of $2x - y$ is:

1. 0

2. 1

3. 3

4. -2

Options 1. 1

2. 2

3. 3

4. 4

Q.13

Suppose that A, B and C are matrices of order $m \times n, n \times 5$ and $5 \times q$ respectively. The restriction on m, n and q so that

$AB - BC$ is defined are

1. n is arbitrary, $q = 3$

2. $m = n, q = 5$
3. $q = 5, m = 3$
4. q is arbitrary, $n = 3$

Options

1. 1
2. 2
3. 3
4. 4

Q.14

If $xy = e^{(x-y)}$, then $\frac{dy}{dx}$ is equal to:

1. $\frac{e^{x-y} + y}{x + e^{x-y}}$
2. $\frac{e^{x-y} - y}{x + e^{x-y}}$
3. $\frac{e^{x-y} - y}{x - e^{x-y}}$
4. $\frac{e^{x-y} + y}{x - e^{x-y}}$

Options

1. 1
2. 2
3. 3
4. 4

Q.15

If $A = [a_{ij}]$ is a square matrix of order 2 such that, $a_{ij} = \begin{cases} 2, & \text{when } i \neq j \\ 0, & \text{when } i = j \end{cases}$, then $\det(A^2)$ is:

- 1.. 0
2. 2
3. 8
4. 16

Options

1. 1
2. 2
3. 3
4. 4

Section : Applied Mathematics

Q.16

In a 500 metres race, the ratio of speeds of two participants Meena and Kamal is 4 : 5 respectively.

If Meena has a start of 200 metres, find the distance by which Meena wins.

1. 375 metres
2. 125 metres
3. 150 metres
4. 300 metres

Options

1. 1
2. 2
3. 3
4. 4

Q.17

If A is a square matrix such that $A^2 = A$ then which of the following statements are TRUE?

(Where I is an identity matrix of same order as A)

- (A) $(I + A)^4 = I + 15A$
- (B) $(I + A)^2 = I + 3A$
- (C) $(I + A)^6 = I + 30A$
- (D) $(I + A)^3 = I + 7A$

Choose the correct answer from the options given below:

1. (A) and (B) only
2. (A), (B) and (C) only
3. (A), (C) and (D) only
4. (A), (B) and (D) only

Options

1. 1
2. 2
3. 3
4. 4

Q.18 The system of equations

$$x - 3y - 8z = -10$$

$$2x + 5y + \lambda z = 13$$

$$3x + y - 4z = 0$$

has infinite number of solutions if the value of λ is equal to:

1. 9
2. 5
3. any real number
4. 6

Options

1. 1
2. 2
3. 3
4. 4

Q.19 Let $A = \begin{bmatrix} \frac{1}{3} & 2 \\ 0 & 2x - 3 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 6 \\ 0 & -1 \end{bmatrix}$.

If $AB = I$ (where I is an identity matrix of order 2), then the value of x is:

1. -1
2. 1
3. 0
4. 2

Options

1. 1
2. 2
3. 3
4. 4

Q.20 **Extra close brace or missing open brace**

per litre so that the resulting syrup would be Rs. 180 per litre?

1. 1 : 3
2. 4 : 3
3. 3 : 1
4. 3 : 4

Options

1. 1
2. 2
3. 3
4. 4

Q.21 A man rows a boat such that it covers a distance upstream in twice the time taken by him to cover the same distance downstream.

If the speed of stream is 5 km/h, the speed of boat in still water is:

1. 5 km/h
2. 10 km/h
3. 15 km/h
4. 25 km/h

Options 1. 1

2. 2

3. 3

4. 4

Q.22 A trust invites a deposit of a lump sum amount from an individual so that an annual scholarship of Rs. 5000 is paid. Rate of interest is 5% per annum.

If the scholarship is to start at the end of this year, then the amount needed to deposit to the trust is:

1. Rs. 60,000

2. Rs. 1,00,000

3. Rs. 1,25,000

4. Rs. 1,50,000

Options 1. 1

2. 2

3. 3

4. 4

Q.23 The inverse of the matrix

$$\begin{bmatrix} 4 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 6 \end{bmatrix}$$
 is

1.
$$\begin{bmatrix} -\frac{1}{4} & 0 & 0 \\ 0 & -\frac{1}{5} & 0 \\ 0 & 0 & -\frac{1}{6} \end{bmatrix}$$

2.
$$\begin{bmatrix} -\frac{1}{30} & 0 & 0 \\ 0 & \frac{1}{24} & 0 \\ 0 & 0 & \frac{1}{20} \end{bmatrix}$$

3.
$$\begin{bmatrix} \frac{1}{4} & 0 & 0 \\ 0 & \frac{1}{5} & 0 \\ 0 & 0 & \frac{1}{6} \end{bmatrix}$$

4.
$$\begin{bmatrix} 6 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

Options 1. 1

2. 2

3. 3

4. 4

Q.24 Differentiation of $\frac{x^3}{1-x^3}$ with respect to x^3 is equal to:

1. $\frac{1}{(1-x^3)^2}, x \neq 1$

2. $\frac{1}{(1-x^3)^3}, x \neq 1$

3. $\frac{1}{(1-x^2)^3}, x \neq 1$

4. $\frac{1}{(1-x^2)^2}, x \neq 1$

Options 1. 1

2. 2

3. 3

4. 4

Q.25 For the given values 27, 35, 42, 45, 51, 34, 43; the five yearly moving averages are:

1. 38, 40, 42

2. 40, 41.4, 43

3. 41.5, 42, 45

4. 42, 44, 45.3

Options 1. 1

2. 2

3. 3

4. 4

Q.26 Which of the following are NOT correct about "Sinking Fund"?

(A) It does not have any specific purpose.

(B) It can be used in any emergency.

(C) Any amount, any time can be deposited in it.

(D) It is set up for a particular upcoming expense.

Choose the correct answer from the options given below:

1. (A), (B) and (D) only

2. (A), (B) and (C) only

3. (A), (C) and (D) only

4. (B), (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.27 Pipe A and B can fill a tank in 15 hours and 20 hours respectively.

Pipe C can empty the tank in 25 hours.

Pipes A, B and C work together but pipe C is closed after 10 hours, then the time taken by pipe A and B to fill the remaining tank is:

1. 12 hours

2. 2 hours

3. 8 hours

4. 4 hours

Options 1. 1

2. 2

3. 3

4. 4

Q.28 The amount to which ₹ 5000 will accumulate at the effective rate of 4% for 4 years and 5% for 2 years is:

1. ₹ 8295

2. ₹ 10295

3. ₹ 6448

4. ₹ 7295

Options 1. 1

2. 2

3. 3

4. 4

Q.29 The function $f(x) = \frac{x}{3} + \frac{3}{x}$ is increasing in the interval:

1. $(-\infty, -3) \cup (3, \infty)$

2. $(-3, 0) \cup (0, 3)$

3. $(-\infty, \infty)$

4. $(0, \infty)$

Options 1. 1

2. 2

3. 3

4. 4

Q.30 If the binomial distribution $X \sim B(n, p)$ has mean 3 and variance $\frac{3}{2}$, $(p + q) = 1$, then which of the following is/are TRUE?

(A) $q = \frac{1}{2}, n = 6$

(B) $P(X \leq 5) = \frac{63}{64}, p = \frac{1}{2}$

(C) $q = \frac{1}{3}, p = \frac{2}{3}$

(D) $P(X = 4) = \frac{15}{64}, n = 6$

Choose the correct answer from the options given below:

1. (A) and (B) only
2. (A), (C) and (D) only
3. (A), (B) and (D) only
4. (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.31 The total cost $C(x)$ associated with the production of x units of an item is given by

$C(x) = 0.001x^3 + 0.06x^2 + 20x + 500$

The marginal cost when 10 units are produced is:

1. 21.5
2. 500
3. 31.5
4. 27.5

Options 1. 1

2. 2

3. 3

4. 4

Q.32 The feasible region of a linear programming problem is bounded.

The corresponding objective function is $Z = 3x - 4y$.

The objective function attains:

1. Only maximum value of Z attained in the feasible region
2. Only minimum value of Z attained in the feasible region
3. Both maximum and minimum value of Z attained in the feasible region
4. Either maximum or minimum of Z attained in the feasible region but not both

Options 1. 1

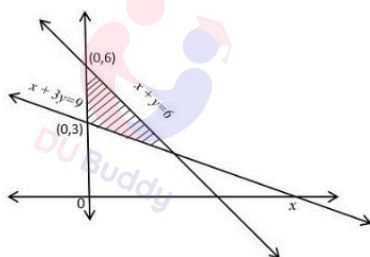
2. 2

3. 3

4. 4

Q.33

The feasible region for an LPP is shown by shaded region in the figure. Then the minimum value of $Z = 11x + 7y$ is



1. 21

2. 42

3. 33

4. 60

Options 1. 1

2. 2

3. 3

Q.34 A charity organization has a fund of Rs. 2,00,000 to provide annual grants to students. The grant amount each year is Rs. 15,000. The fund earns an interest rate of $r\%$ per annum. If the interest earned is used entirely to provide the grants, then the annual interest rate r is:

1. 8%
2. 7.5%
3. 5.7%
4. 9.2%

Options

1. 1
2. 2
3. 3
4. 4

Q.35 Mrs Rathna invested Rs. 2 lakh in an enterprise for 5 years. Her compound annual growth rate (CAGR) turned out to be 20.5%. The end balance would be (given $(1.205)^5 = 2.54$):

1. Rs. 5.08 lakh
2. Rs. 4 lakh
3. Rs. 2.41 lakh
4. Rs. 22.05 lakh

Options

1. 1
2. 2
3. 3
4. 4

Q.36 Consider the following data of expenses (in lakhs) of an organization year wise:

Year	2001	2002	2003	2004	2005
Expenses(Rs. lakh)	160	185	220	300	510

Then expected expenses trend for the year 2006 using method of least squares is:

1. Rs. 419.5 lakh
2. Rs. 620.5 lakh
3. Rs. 319.5 lakh
4. Rs. 519.5 lakh

Options

1. 1
2. 2
3. 3
4. 4

Q.37 Every year the price of a motorcycle depreciates by Rs. 15,000. After 12 years its price has become Rs. 50,000. Then its original cost was:

1. Rs. 1,80,000
2. Rs. 2,15,000
3. Rs. 2,30,000
4. Rs. 2,45,000

Options

1. 1
2. 2
3. 3
4. 4

Q.38 A random variable X follows Poisson distribution such that $P(X = 1) = 2P(X = 2)$, then $P(X = 0)$ is:

1. e

2. $\frac{1}{e}$
3. 1
4. e^2

Options 1. 1

2. 2
3. 3
4. 4

Q.39 The supply function of a commodity is $P = x^3 + 2x + 18$.
When 4 units of the commodity are sold, then producer surplus is:

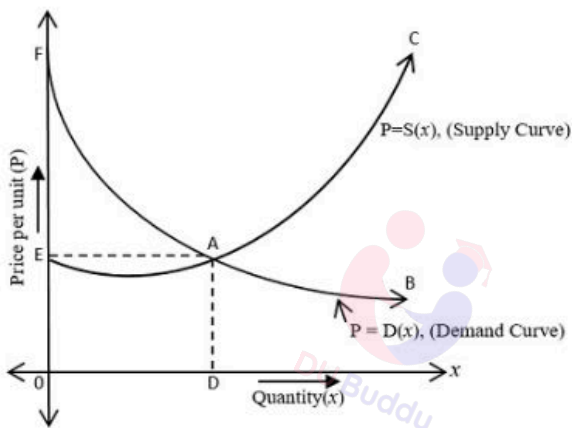
1. 320
2. 488
3. 160
4. 208

Options 1. 1

2. 2
3. 3
4. 4

Q.40

In the below mentioned demand supply curve, identify the equilibrium point



1. C
2. B
3. A
4. F

Options 1. 1

2. 2
3. 3
4. 4

Q.41 Rakshita plans to buy a house for Rs. 1,00,00,000 with down payment of 20% of the value of the house paid by her mother. Rest of the amount she wishes to pay in 25 years by equal monthly instalments at an interest of 9% per annum compounded monthly. Then the EMI paid by her is (Given $(1.0075)^{300} = 9$):

1. Rs. 50,000
2. Rs. 40,000
3. Rs. 70,700
4. Rs. 67,500

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.42 A die is thrown twice. The probability of occurrence of 5 at least once is:

- 1. $\frac{11}{36}$
- 2. $\frac{7}{12}$
- 3. $\frac{35}{36}$
- 4. $\frac{20}{36}$

Options

- 1. 1
- 2. 2
- 3. 3
- 4. 4

Q.43 Consider the following hypothesis test:

$$H_0 : \mu = 20$$

$$H_a : \mu \neq 20$$

A sample of 40 provided a sample mean of 19.

The standard deviation is 3.

Then the value of the t-test statistic is:

- 1. - 1.8
- 2. - 2.1
- 3. 2.1
- 4. 1.8

Options

- 1. 1
- 2. 2
- 3. 3
- 4. 4

Q.44 If a sample has n observations $x_1, x_2, \dots, x_n$ with m constraints on these values, then the degree of freedom of the sample statistic is:

- 1. $\frac{n - m}{2}$
- 2. $\frac{n + m}{2}$
- 3. $n + m$
- 4. $n - m$

Options

- 1. 1
- 2. 2
- 3. 3
- 4. 4

Q.45 The rise in prices before Diwali is an example of:

- 1. Seasonal trend
- 2. Cyclical trend
- 3. Long term trend
- 4. Irregular trend

Options

- 1. 1
- 2. 2
- 3. 3
- 4. 4

Q.46

The total cost function is given by $C(x) = \frac{1}{3}x^3 - 5x^2 + 30x - 15$ and selling price per unit is Rs. 6.

The profit is maximum if the value of x is:

1. $x = 4$
2. $x = 6$
3. $x = 5$
4. $x = 10$

Options 1. 1

2. 2
3. 3
4. 4

Q.47

If $\frac{1}{|x| - 3} \leq \frac{1}{2}$, then the value of x is:

1. $x \in (-\infty, -6) \cup (-3, 3) \cup (5, \infty)$
2. $x \in (-\infty, 5) \cup (5, \infty)$
3. $x \in (-\infty, -3) \cup (3, \infty)$
4. $x \in (-\infty, -5] \cup (-3, 3) \cup [5, \infty)$

Options 1. 1

2. 2
3. 3
4. 4

Q.48

If $x = \frac{a}{1+t}$ and $y = \frac{a}{(1+t)^2}$ where $a > 0$,

then $\frac{d^2y}{dx^2}$ at $t = 1$ is:

1. $\frac{1}{a}$
2. $\frac{2}{a}$
3. $\frac{4}{a}$
4. Not defined

Options 1. 1

2. 2
3. 3
4. 4

Q.49

If matrix $A_p = \begin{bmatrix} p & (p+1) \\ p & (p-1) \end{bmatrix}$, $p \in \mathbb{N}$ (where $\mathbb{N}$ is the set of natural numbers),

then the value of $|A_1| + |A_2| + |A_3| + \dots + |A_{2025}|$ is:

1. $-(2025)(2026)$
2. $(2025)^2$
3. $(2025)(2026)$
4. $-(2025)^2$

Options 1. 1

2. 2
3. 3
4. 4

Q.50

A 95% confidence interval for a population mean was reported to be 152 to 160.

If the standard deviation $\sigma = 15$,

then the sample size is (given $Z_{0.025} = 1.96$) :

1. 40
2. 60

4. 50

2.2

4.4

1

Section : Core Mathematics

Q.51 $\frac{d^2}{dx^2} \left(\det \begin{bmatrix} x^3 & x \\ 2 & e^x \end{bmatrix} \right)$ equals

1. $(xe^x(x^2 - 6x + 6))$
2. $(x^2e^x(x^2 + 6x + 6))$
3. $(xe^x(x^2 + 6x + 6))$
4. $(e^x(x^2 - 6x + 6))$

Options 1. 1

3.3

Match List-I with List-II

If the random variable x has the following distribution:

x	0	1	2	otherwise
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$$P(x) \quad k \quad k \quad 2k \quad 0$$

List-I & List-II

List-I

List-II

(A) k

$$(I)^{\frac{3}{4}}$$
$$\mathbb{I}(B); P(x \geq 2) \quad (II) \quad 1/4$$

(II) 1/4

$$(C) P(x \leq 2) \quad (III) \frac{1}{2}$$
$$(III) \frac{1}{2}$$

(D) $P(0 < x \leq 2)$ (IV) 1

1) (IV) 1

Choose the correct answer from the options given below:

1. $(A) - (I), (B) - (III), (C) - (II), (D) - (IV)$
2. $(A) - (II), (B) - (III), (C) - (I), (D) - (IV)$
3. $(A) - (I), (B) - (IV), (C) - (III), (D) - (II)$
4. $(A) - (II), (B) - (III), (C) - (IV), (D) - (I)$

Options 1. 1

3.3

Then the objective function attains:

1. Only maximum in the feasible region

2. Only minimum in the feasible region
3. both maximum and minimum in the feasible region
4. either maximum or minimum but not both in the feasible region

Options

1. 1
2. 2
3. 3
4. 4

Q.54

If $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$, then $A^2 - 5A$ is equal to (where I is identity matrix of order 2)

1. $14I$
2. $7I$
3. $-7I$
4. $-5I$

Options

1. 1
2. 2
3. 3
4. 4

Q.55

The area of the region bounded by the curves $y = x^2 + 2$, $y = x$, $x = 0$ and $x = 2$ is

1. $\frac{11}{3} \text{ Sq. units}$
2. $\frac{2}{3} \text{ Sq. units}$
3. $\frac{14}{3} \text{ Sq. units}$
4. $\frac{26}{3} \text{ Sq. units}$

Options

1. 1
2. 2
3. 3
4. 4

Q.56

The value of $\cos(2 \cos^{-1} x + \sin^{-1} x)$ at $x = \frac{1}{5}$ is

1. $\sqrt{\frac{25}{24}}$
2. $-\sqrt{\frac{24}{25}}$
3. $\sqrt{\frac{24}{25}}$
4. $-\sqrt{\frac{25}{24}}$

Options

1. 1
2. 2
3. 3
4. 4

Q.57

If $\vec{a} \cdot \vec{b} = \vec{a} \cdot \vec{c}$, $\vec{a} \times \vec{b} = \vec{a} \times \vec{c}$ and $\vec{a} \neq \vec{0}$, then the vector $\vec{b}$ is equal to:

1. $\vec{0}$

2. $\vec{c}$
3. $\vec{a}$
4. $(\vec{a} + \vec{c})$

Options 1. 1
2. 2
3. 3
4. 4

Q.58

If $A = \begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix}$ and $f(x) = 2x^2 - 4x + 5$, then $f(A)$ is equal to

1. $\begin{bmatrix} 19 & -16 \\ -32 & 51 \end{bmatrix}$
2. $\begin{bmatrix} 19 & -32 \\ -16 & 51 \end{bmatrix}$
3. $\begin{bmatrix} 19 & -11 \\ -27 & 51 \end{bmatrix}$
4. $\begin{bmatrix} -19 & 16 \\ 32 & -51 \end{bmatrix}$

Options 1. 1
2. 2
3. 3
4. 4

Q.59

For the differential equation $x \frac{dy}{dx} + 3y = x^2 \log_e x$, which of the following statements are TRUE?

- (A) Product of order and degree is 1
- (B) Integrating factor is x^3
- (C) Integrating factor is $3x$
- (D) General solution is $y = \frac{x^3}{36} (6 \log_e |x| - 1) + Cx^{-3}$, C is an arbitrary constant

Choose the correct answer from the options given below:

1. (A), (C) and (D) only
2. (A) and (B) only
3. (A) and (D) only
4. (A), (B) and (D) only

Options 1. 1
2. 2
3. 3
4. 4

Q.60

The direction cosines of a line which makes equal angles with co-ordinate axes are:

1. $\pm \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right)$
2. $\pm (1, 1, 1)$
3. $\pm \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3} \right)$
4. $\pm \left(\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}} \right)$

Options 1. 1
2. 2
3. 3
4. 4

Q.61

The function $f(x) = x^2 - x + 1$ is

1. Increasing on $\left(-\frac{1}{2}, 1\right)$ and decreasing on $\left(0, \frac{1}{2}\right)$
2. Increasing on $\left(\frac{1}{2}, \infty\right)$ and decreasing on $\left(-\infty, \frac{1}{2}\right)$
3. Increasing on $\left(-\infty, \frac{1}{2}\right]$ and decreasing on $\left[\frac{1}{2}, \infty\right)$
4. Increasing on $(-\infty, 1)$ and decreasing on $(1, \infty)$

Options 1. 1

2. 2
3. 3
4. 4

Q.62 The number of arbitrary constants in the general solution of a differential equation of order 4 and degree 1 is

1. 0
2. 4
3. 2
4. 1

Options 1. 1

2. 2
3. 3
4. 4

Q.63 The function $f : [-1, 1] \rightarrow \mathbb{R}$ (set of real numbers) given by $f(x) = \frac{x}{x+3}$ is

1. one-one only
2. onto only
3. both one-one and onto
4. neither one-one nor onto

Options 1. 1

2. 2
3. 3
4. 4

Q.64 Assume that R is a relation on the set $\mathbb{Z}$ of integers and it is given by $(x, y) \in R \Leftrightarrow |x - y| \leq 1$. Then, R is

1. Reflexive and symmetric but not transitive
2. An equivalence relation
3. Symmetric and transitive but not reflexive
4. Reflexive and transitive but not symmetric

Options 1. 1

2. 2
3. 3
4. 4

Q.65 If $f(x) = |x| + |x - 5|$, then which of the following statements are TRUE?

- (A) f is a continuous function everywhere
- (B) f is a continuous function except $x = 5$ and $x = 0$
- (C) f is a continuous function except $x = 0$ but not differentiable at $x = 5$
- (D) f is a continuous function everywhere but not differentiable at $x = 0$ and $x = 5$

Choose the correct answer from the options given below:

1. (A) and (B) only
2. (B) and (C) only
3. (A) and (D) only
4. (B) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.66 If $\vec{a}$ and $\vec{b}$ are two non-zero orthogonal vectors, then $|\vec{a} + \vec{b}|$ is equal to:

1. 0
2. $2|\vec{b}|$
3. $2|\vec{a}|$
4. $|\vec{a} - \vec{b}|$

- Options 1. 1
2. 2
3. 3
4. 4

Q.67 A bag contains 8 blue and 12 green balls. Two balls are drawn in succession without replacement.

The probability that first is blue and second is green is

1. $\frac{42}{95}$
2. $\frac{71}{95}$
3. $\frac{8}{25}$
4. $\frac{24}{95}$

- Options 1. 1
2. 2
3. 3
4. 4

Q.68 Let the equations of lines be $L_1 : \vec{r}_1 = \vec{a}_1 + \lambda \vec{b}_1$ and $L_2 : \vec{r}_2 = \vec{a}_2 + \lambda \vec{b}_2$

such that $\vec{a}_1 - \vec{a}_2 = 2\vec{i} + 4\vec{j} + 4\vec{k}$ and $\vec{b}_1 \times \vec{b}_2 = 8\vec{i} - 4\vec{k}$.

Then the shortest distance between L_1 and L_2 is

1. 20
2. 4
3. 0
4. $2\sqrt{3}$

- Options 1. 1
2. 2
3. 3
4. 4

Q.69 Match List-I with List-II

List-I

List-II

Integral

Solution: C is an arbitrary constant

(A) $\int \frac{dx}{x^2 + 25}$ (I) $\frac{1}{10} \log \left| \frac{5+x}{5-x} \right| + C$

(B) $\int \frac{dx}{x^2 - 25}$ (II) $\log |x + \sqrt{x^2 - 25}| + C$

(C) $\int \frac{dx}{25 - x^2}$ (III) $\frac{1}{5} \tan^{-1} \left(\frac{x}{5} \right) + C$

$$(D) \int \frac{dx}{\sqrt{x^2 - 25}} \quad (IV) \quad \frac{1}{10} \log \left| \frac{x-5}{x+5} \right| + C$$

Choose the correct answer from the options given below:

1. (A) – (I), (B) – (IV), (C) – (II), (D) – (III)
2. (A) – (III), (B) – (IV), (C) – (I), (D) – (II)
3. (A) – (III), (B) – (IV), (C) – (II), (D) – (I)
4. (A) – (II), (B) – (IV), (C) – (III), (D) – (I)

Options

1. 1
2. 2
3. 3
4. 4

Q.70 The rate of change of volume of a sphere with respect to its surface area, when the radius is 6 cm is:

1. $1 \text{ cm}^3/\text{cm}^2$
2. $6 \text{ cm}^3/\text{cm}^2$
3. $2 \text{ cm}^3/\text{cm}^2$
4. $3 \text{ cm}^3/\text{cm}^2$

Options

1. 1
2. 2
3. 3
4. 4

Q.71 If $\int_0^a 3x^2 dx = 8$, then the value of a is:

1. 2
2. 10
3. 8
4. 5

Options

1. 1
2. 2
3. 3
4. 4

Q.72 If the system of equation

$$x - y + z = 4$$

$$x - 2y - 2z = 9$$

$$2x + y + \lambda z = 1$$

has a unique solution, then

1. $\lambda \geq 11$
2. $\lambda = 11$
3. $\lambda \neq 11$
4. $\lambda \leq 11$

Options

1. 1
2. 2
3. 3
4. 4

Q.73 A vector of magnitude 9, which is perpendicular to both the vectors $(4\hat{i} - \hat{j} + 8\hat{k})$ and $(-\hat{j} + \hat{k})$ is

1. $7\hat{i} - 4\hat{j} - 4\hat{k}$
2. $4\hat{i} - 7\hat{j} + 4\hat{k}$
3. $4\hat{i} - 4\hat{j} + 7\hat{k}$
4. $-4\hat{i} + 4\hat{j} - 7\hat{k}$

Options 1. 1

2. 2

3. 3

4. 4

Q.74 Let A and B be two events. Then which of the following statements are TRUE?

(A) $P(B | A) = \frac{P(A \cap B)}{P(A)}$, provided $P(A) \neq 0$

(B) $P(B') = 1 + P(B)$

(C) $P(A \cup B) = P(A) + P(B) + P(A \cap B)$

(D) $P(A \cap B) = P(A) \cdot P(B)$ if A and B are independent events

Choose the correct answer from the options given below:

1. (A), (B) and (C) only

2. (B), (C) and (D) only

3. (A) and (D) only

4. (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.75 The area (in sq. units) of the bigger portion of region enclosed by the curves $4x^2 + 9y^2 = 36$ and $2x + 3y = 6$ is

1. $\frac{3}{2}(\pi - 2)$

2. $\frac{3}{2}(3\pi + 2)$

3. (6π)

4. $(3\pi + 3)$

Options 1. 1

2. 2

3. 3

4. 4

Q.76 The corner points of the feasible region with the constraints $x + y \leq 30$, $x + y \geq 15$, $y \leq 20$, $x \leq 15$, and $x, y \geq 0$ are

1. (0, 0), (0, 15), (0, 20), (15, 0), (10, 20)

2. (0, 15), (20, 0), (15, 0), (10, 20)

3. (15, 0), (15, 15), (10, 20), (0, 20), (0, 15)

4. (0, 0), (20, 0), (0, 15), (15, 15)

Options 1. 1

2. 2

3. 3

4. 4

Q.77 $\int \frac{dx}{(1 + 5 \sin^2 x)}$ is equal to

1. $\sqrt{6} \cot^{-1}(\sqrt{6} \cot x) + C$: C is an arbitrary constant

2. $\sqrt{6} \cot^{-1}(\sqrt{6} \tan x) + C$: C is an arbitrary constant

3. $\frac{1}{\sqrt{6}} \tan^{-1}(\sqrt{6} \cot x) + C$: C is an arbitrary constant

4. $\frac{1}{\sqrt{6}} \tan^{-1}(\sqrt{6} \tan x) + C$: C is an arbitrary constant

Options 1. 1

2. 2

3. 3

Q.78 If the area of a triangle whose vertices are $(-1, 3)$, $(1, -5)$ and $(k, 2)$ where $k > 0$ is 30 sq. units, then the value of k is

1. 4
2. 27
3. $\frac{4}{27}$
4. $\frac{27}{4}$

Options 1. 1

2. 2
3. 3
4. 4

Q.79

If $x \neq y \neq z$ then $\begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix}$ is equal to

1. $x + y + z$
2. xyz
3. $(x - y)(y - z)(z - x)$
4. 0

Options 1. 1

2. 2
3. 3
4. 4

Q.80 Which of the following statement are correct?

- (A) $A = [a_{ij}]_{n \times n}$ is a diagonal matrix if $a_{ij} = 0$ when $i \neq j$
- (B) A square matrix $A = [a_{ij}]$ is called a symmetric matrix if $a_{ij} = a_{ji}$ for all i, j
- (C) A square matrix $A = [a_{ij}]$ is called a skew-symmetric matrix if $a_{ij} = -a_{ji}$ for all i, j
- (D) For every square matrix A , there exist an identity matrix of the same order such that $IA = AI = I$

Choose the correct answer from the options given below:

1. (A), (B) and (C) only
2. (B), (C) and (D) only
3. (A) and (C) only
4. (B) and (C) only

Options 1. 1

2. 2
3. 3
4. 4

Q.81 Let $f(x) = 4x^3 - 18x^2 + 27x - 5$, $x \in \mathbb{R}$. Then which of the following statements are TRUE?

- (A) $f''(x) = 24x - 36$
- (B) f has local maxima at $x = \frac{3}{2}$ but no minima
- (C) f has neither maxima nor minima
- (D) f has both maxima and minima

Choose the correct answer from the options given below:

1. (A) and (B) only
2. (A) and (C) only
3. (B) and (C) only
4. (A) and (D) only

Options 1. 1

2. 2
3. 3

Q.82 Match List-I with List-II

Let θ be the angle between the vector $\vec{a}$ and $\vec{b}$.

List-I

(A) $\vec{a} \cdot \vec{b}$

(B) $\vec{a} \times \vec{b}$

(C) Projection vector of $\vec{a}$ on $\vec{b} (\neq 0)$ (D) $\vec{a}$ and $\vec{b}$ are orthogonal vectors

List-II

(I) $\frac{\vec{a} \cdot \vec{b} \vec{b}}{|\vec{b}|^2}$

(II) $\vec{a} \cdot \vec{b} = 0$

(III) $|\vec{a}||\vec{b}| \sin \theta \hat{n}$ where $\hat{n}$ is a unit vector perpendicular to both $\vec{a}$ and $\vec{b}$

(IV) $|\vec{a}||\vec{b}| \cos \theta$

Choose the correct answer from the options given below:

1. (A) – (I), (B) – (II), (C) – (III), (D) – (IV)
2. (A) – (IV), (B) – (I), (C) – (III), (D) – (II)
3. (A) – (IV), (B) – (III), (C) – (I), (D) – (II)
4. (A) – (II), (B) – (IV), (C) – (I), (D) – (III)

Options

1. 1
2. 2
3. 3
4. 4

Q.83 The probabilities of occurrence of two events A and B are 0.45 and 0.20 respectively.

The probability of their simultaneous occurrence is 0.06.

The probability that neither A nor B occurs is

1. 0
2. 1
3. 0.41
4. 0.59

Options

1. 1
2. 2
3. 3
4. 4

Q.84

If $x = a \sin 2t(1 + \cos 2t)$ and $y = b \cos 2t(1 - \cos 2t)$, then $\left(\frac{dy}{dx}\right)_{x=\frac{\pi}{4}}$ is equal to

1. ab
2. $\frac{b}{a}$
3. $\frac{a}{b}$
4. $\frac{1}{ab}$

Options

1. 1
2. 2
3. 3
4. 4

Q.85 The vector equation of line passing through $(2, -1, 3)$ and perpendicular to the lines

$$\frac{x-2}{3} = \frac{y-1}{1} = \frac{z+2}{2} \quad \text{and} \quad \frac{x+3}{-4} = \frac{y-5}{-3} = \frac{z+1}{2}$$

(Here λ is a parameter)

$$1. \vec{r} = (2\hat{i} - \hat{j} + 3\hat{k}) + \lambda(8\hat{i} - 14\hat{j} - 5\hat{k})$$

$$2. \vec{r} = (-2\hat{i} + \hat{j} - 3\hat{k}) + \lambda(8\hat{i} - 14\hat{j} - 5\hat{k})$$

$$3. \vec{r} = (-8\hat{i} - 14\hat{j} - 5\hat{k}) + \lambda(-2\hat{i} + \hat{j} + 3\hat{k})$$

$$4. \vec{r} = (8\hat{i} - 14\hat{j} - 5\hat{k}) + \lambda(2\hat{i} - \hat{j} + 3\hat{k})$$

Options 1. 1

2. 2

3. 3

4. 4